

Theory 1

1. In relation to events in Java Swing...
 - a. We can associate more than one event handler of the same type to the same source object **True**
 - b. We can associate more than one event handler of the different type to the same source object **True**
 - c. We can share the same event handler instance with more than one source objects of different types, whenever the code they execute is compatible with both source object types **True**
 - d. A and B are True **True**
 - e. All of them are True **True**
 2. Considering the code... Which of the following sentences is right?
 - a. The method is right and the background of the source object will be changed to red **False**
 - b. The method is incorrect, some parameters are missing in the method getSource (line 2) **False: setSource is correct**
 - c. The method is incorrect because a parameter must be added in order to know the area that must be painted **False**
 - d. The method is incorrect because area should be of type Component **False**
 - e. None of the answers is right **True**
- ```
1 public void paint (KeyEvent e){
2 JTextArea area = (Component) e.getSource(); doesn't compile for every component, because is a
3 area.setBackground(Color.red);
4 }
```
3. Which of the following would be the right sentence if we want to register a listener object (objReceiver) and a source (objSource) object for a mouse event?
    - a. objSource.addMouseAdapter(objSource) **False**
    - b. objSource.addMouseListener(objReceiver) **True**
    - c. objSource.addMouseEvent(objReceiver) **False**
    - d. The answer A if we create the listener extending the adapter class and B if we directly implement the listener interface **False**
    - e. None of them is right **False**
  4. The listener class that manages a KeyListener (consider that the listener interface contents more than one method!)...
    - a. Can be built extending the KeyListener interface **False: implements**
    - b. Can be built extending the adapter class KeyAdapter **True**
    - c. Can be built extending the KeyAdapter interface **False: not interface**
    - d. (a) and (b) are both suitable **False**
    - e. None of the answers is right **False**
  5. Given the code in the frame and considering that the interface KeyListener includes more than 2 methods...
    - a. pT represents the source object of the event and area1 represents the listener object **False (opposite)**
    - b. The instruction in line 16 is wrong, given that it should be area1.addKeyAdapter(pT); **False**
    - c. The instruction in line 5 is wrong, given that we should use extends instead of implements **False**
    - d. The class ProcesaTecla is wrong because every method of the KeyListener class should had been implemented **True (implements Listener)**
    - e. More than one answer is right **False**

```
1 public class MainWindow extends JFrame {
2 private JTextArea area1 = null;
3 private ProcessKey pT = null;
4 ...
5 class ProcessKey implements KeyListener {
```

```

6 public void keyTyped (KeyEvent e){
7 printar(e);
8 }
9 }
10 private void initialize(){
11 pT = new ProcessKey();
12 ...
13 }
14 private void getArea1{ non sense
15 ...
16 area 1.addKeyListener(p1);
17 }
18 }

```

6. This image corresponds to a capture of an application that allows you to manage television. What principle/s of usability have been applied?
  - a. Response time **False**
  - b. Familiarity **True: metaphors**
  - c. Recoverability **False: maybe if it has a change channel, so you can recover a before step**
  - d. More than one answer is correct **False**
7. Regarding Swing and AWT it's true that:
  - a. Swing is a library that improves and extends the components provided by AWT
  - b. AWT and Swing are part of JFC (Java Foundation Classes)
  - c. In AWT there is an event library to carry out event management
  - d. The above 3 answers are true **True**
8. The checkboxes...
  - a. Must be used to represent exclusive options **False: you can do it, but you shouldn't**
  - b. They should be used to represent non-exclusive options **True**
  - c. They can never be used in menus as they do not support image **False: we can use them in menus, but it's impractical because it closes after clicking one, so you have to go again**
  - d. Answers a and b are true **False**
9. A sequence of phases in the cyclical process of UCD, starting from an outline of initial requirements, would be:
  - a. Specify the Context of Use - Evaluate- Produce Design Solutions - Evaluate **False: you cannot produce design solutions before specifying requirements**
  - b. Specify the Context of Use - Specify Requirements - Produce Design Solutions - Evaluate **True**
  - c. Specify Requirements - Produce Design Solutions - Specify the Context of Use - Evaluate **False**
  - d. Evaluate **False**
  - e. All answers are true **False**
10. What of the following statements is true?
  - a. A label must keep enabled even when the component it's associated to is disabled **False**
  - b. A label must be disabled when the component it's associated to is disabled **True**
  - c. If we want to provide a keyboard access to a component that do not accept it directly (as a textfield, for example), we must modify the label for property of that component **False: label for is changed in the label**
  - d. A and C are true **False**
11. These 3 snapshots belong to 3 different applications that show the shopping cart in the upper-right corner of the view. Which principles of usability have been applied?
  - a. Recoverability **False: system doesn't crash**
  - b. Consistency **True: is in the same corner as all the apps do**

- c. Familiarity **True: shopping cart**
- d. More than one is applied **True: B and C**

**AB test: show that users have difficulties or a step is longer in UCD. So you make 2 apps to compare (change of a color, icon, position in a button)**

12. In the User Centered Design...

- a. It's suitable only for the development of desktop applications **False**
- b. It's based in the iterative repetition of its phases until we reach a product that satisfies user's needs and requirements **True**
- c. It has 4 sequential phases that are executed once each of them **False: it's a cycle**
- d. More than one answer is right **False**

13. Which of the following contributes to increase the usability of an application?

- a. The user's sensorial memory capacity when using an application **False**
- b. The fact that notification of changes in the system will not be immediate **False: the opposite (principle of synthesis)**
- c. Updating frequently the UI with any new innovative interaction technique we discover **False (makes users crazy. Test it before)**
- d. The use of metaphors of the real world **True**
- e. More than one answer is true **False**

14. In relation to tooltips:

- a. They ease the burden of long-term memory **True**
- b. Provide keyboard access **False: mnemonics**
- c. It's recommended that there is an option to disable them **True: to don't overload the system**
- d. More than one answer is true **True**

15. The comboBox in the Image that has the text not ordered, and has a button Go next to it

- a. Satisfies every content recommendation **False: content must be ordered**
- b. It doesn't need keyboard access as it's accompanied by a button ("Go") **False**
- c. Doesn't comply with every content recommendation **True**
- d. More than one answer is true **False**
- e. None of the answers is true **False**

16. Regarding the Persona-Scenario technique, we can state that:

- a. Provide archetypes of people that will use the application **True**
- b. A minimum of 8 characters is recommended to increase the reliability of the technique **False (maximum 7)**
- c. Doesn't include user behaviour, this is considered in the evaluation phase with real users **False (scenarios = behaviour)**
- d. More than one answer is right **False**

17. Fitts Law:

- a. It estimates the time invested in making a decision **False: Hicks**
- b. Consider that, if the same action is repeated over time, performance tends to improve **False: law of practice**
- c. It expresses that the time required to reach a visual objective is a function of the distance and size **True**
- d. Defines a quantitative method that allows the validation of an interface in terms of the time that the user invests in performing a task **False**

18. Which of the following statements is true?

- a. The user models based on biology are based on interviews with users to extract their common feature **False: Personas (cultural)**

- b. The objective of user models is to predict the behaviour of the users they represent **True**
- c. The time that the user spends moving his hand from one device to another (from keyboard to mouse, for example) is calculated in GOMS applying Fitts' law **False: GOMS assumes that everybody will be witch each device and we don't know distances**
- d. The time the user needs to select an option from a menu is independent of the number of options on the menu **False: Hicks**

## Theory 2

1. Regarding the components that allow the user to make a selection, it's true that:
  - a. The lists (JList) allow selecting one or more items from the list, depending on the value of one of its attributes. **True**
  - b. A FileChooser allows selecting one or several files, depending on the value of one of its attributes. **True: enableMultipleSelectionOption()**
  - c. ComboBoxes (JComboBox) allow you to select one or more items from the combo, depending on the value of one of its attributes. **False**
  - d. More than one answer is true. **True**
2. We start from a help support system already built with JavaHelp in which there is a table of contents, an index and the search view. We need to modify the html file that is displayed by default, where the application is fully described (Introduction.html), adding more text. What do you need to do so that the new text entered is updated in the 3 views?

**Map file is the same (you don't add any file)**

**You don't change the default page, so you don't need to change help set**

- a. Modify the map file and run jhindexer **False**
  - b. Modify the map, toc, index files and execute jhindexer **False**
  - c. Modify all the help xml files, including the helpset, and run jhindexer **False**
  - d. None of the answers is correct. **True. Maybe modify index if we are adding more items, but we have to run jhindexer**

3. Given this code and taking into account that:

The code corresponding to the Book class is found in the file books.jar located in the directory C:\utilities

The application (Library.class) is in the directory C:\Program files\applications\library

```
class Library {
 public static void main (String[] args){
 ...
 Book book = new Book();
 book.setTitulo("Aquitaine");
 ...
 }
}
```

Indicate the correct value for the Classpath environment variable

**Classpaths have to point to every root of the fully qualified name of classes. We need to include .jar in the classpath: c:\utilities\books.jar**

**Since there is no package, the class path must point to the package.**

**If it had before class Library the package library, then the correct one is a**

- a. Classpath = c:\Program Files\applications; c:\utilities\books.jar **False**
  - b. Classpath = c:\Program Files\applications\library; c:\utilities **False**
  - c. Classpath = c:\Program Files\applications\library; c:\utilities\books.jar **True**
  - d. Classpath = c:\Program Files\applications; c:\utilities **False**
4. Suppose that in the previous example the application is localized for Spanish (es) and English (en). In case you want to localize for French (fr) we would have to:
  - a. Modify the file Texts\_es\_en.properties adding the new text in French and change their name to Texts\_es\_en\_fr.properties **False**
  - b. Add a new line to the code System.out.println(messages.getString("t3")); **False**
  - c. Create a new Texts\_fr.properties **True**
  - d. More than one of the answers is true **False**

5. The constraints property of a component:

- a. In components such as combobox, define the array of elements to be displayed through it **False. ComboBox is in model**
  - b. Refers to the position of the component with respect to a layout **True -> North, south....**

- c. The component has to be a container and the property refers to the array of components added to it **False**
- d. The component has to be text component and the property refers to the number of characters added to it **False**

6. The Card Sorting Technique:

- a. It's a user experience design technique that is used to classify users, and thus, improve both the design and the results obtained in the test plan **False**
- b. Aims to find a logical and simple way to organize the contents of an application so that everything is where users expect it to find it **True**
- c. It allows to represent by means of cards a set of potential users randomly selected from the target audience **False**
- d. More than one of the above answers is true **False**